

J McCann & Co Ltd · KTP Associate Task

Deliverable 1

Requirements Specification

Backpack — a mobile and web platform enabling on-site operatives to log job activity and giving the Quantity Surveying team accurate, real-time cost intelligence.

PROJECT	Backpack Prototype
DOCUMENT	1 — Requirements List
VERSION	1.0
PREPARED FOR	Dr N. Caldwell, University of Suffolk

1. Introduction

This document defines the functional, non-functional and constraint-based requirements for the prototype of Backpack — a software system commissioned by J McCann & Co Ltd to digitise the field-to-finance workflow of its highway lighting and electrical operations across the East of England.

Today, on-site activity is recorded on paper and re-keyed into spreadsheets by office staff, creating delays, data loss and disputed costings. Backpack replaces this with a mobile-first capture tool for operatives and a connected web dashboard for the Quantity Surveying (QS) team, with automatic cost computation as the single source of truth.

1.1 Purpose

To establish a clear, prioritised and testable set of requirements that scope the prototype, frame architectural decisions, and provide a baseline for evaluation at interview and beyond.

1.2 Scope of the Prototype

The prototype delivers an end-to-end vertical slice: operative login, a single fully-functional job-entry screen, mock cost rate cards, and a costed job summary visible to the QS role. The full production system is described where it informs architectural choices but is not implemented in code.

1.3 Stakeholders

- Field Operatives — primary users of the mobile app; capture site activity.
- Quantity Surveyors — primary users of the web dashboard; review and approve costs.
- Operations Managers — secondary users; oversight, analytics and exceptions.
- J McCann IT & KTP Academic Supervisors — governance, technical assurance, KTP outcomes.

1.4 Requirement Prioritisation (MoSCoW)

Each requirement is tagged Must, Should or Could. Must requirements are in scope of the prototype; Should requirements are designed-for but may be stubbed; Could requirements are documented for the production roadmap.

2. Functional Requirements

Functional requirements describe what the system must do. They are grouped by user role and by the cross-cutting cost-calculation engine that binds the two roles together.

2.1 Operative — Mobile App

The operative is the system's primary data source. The mobile app is designed for use in the field, often in adverse conditions, frequently with gloves, and not always with reliable network coverage.

ID	Requirement	Priority	Acceptance Criteria
FR-OP-01	Secure user authentication with role-based access	Must	Operative can sign in with credentials; session persists until explicit sign-out; role determines visible screens.
FR-OP-02	Select or scan a job reference to begin a record	Must	Job can be chosen from a list of assigned jobs or by scanning a QR/barcode; the selected reference is bound to the record.
FR-OP-03	Log arrival time on-site	Must	Timestamp captured with one tap; editable before submission; defaults to current device time.
FR-OP-04	Log departure time from site	Must	Departure cannot precede arrival; system computes elapsed on-site duration in minutes.
FR-OP-05	Record plant and tools used	Must	Multi-select from a predefined list with hourly rates; each entry has a duration of use.
FR-OP-06	Record materials consumed with quantities	Must	User adds line items {material, quantity}; quantities validated as positive numbers.
FR-OP-07	Record vehicles on-site with arrival time	Must	User selects vehicle(s) from fleet list; arrival time stored per vehicle.
FR-OP-08	Capture free-text notes against a job	Should	Multi-line text field, max 2,000 characters; saved with the record.
FR-OP-09	Capture photographic evidence of completed work	Should	Camera and gallery upload; images compressed client-side; thumbnails shown before submit.
FR-OP-10	Submit completed job record	Must	On submit, the record is validated, posted to the backend, and a confirmation is shown with a record ID.
FR-OP-11	View history of previously submitted jobs	Should	Chronological list filterable by date; each item links to a read-only detail view.

ID	Requirement	Priority	Acceptance Criteria
FR-OP-12	Operate without network connectivity	Must	Records can be created and queued offline; queued records sync automatically when connectivity returns; conflicts surfaced to user.

2.2 Quantity Surveyor — Web Dashboard

The QS consumes the data produced in the field. The dashboard turns raw site activity into priced, auditable, exportable records that feed the company's commercial and finance processes.

ID	Requirement	Priority	Acceptance Criteria
FR-QS-01	View all submitted job records	Must	Paginated list with key fields visible (ref, operative, date, total cost, status).
FR-QS-02	View itemised cost breakdown per job	Must	Detail screen shows labour, plant, materials, vehicle subtotals and a grand total.
FR-QS-03	Filter and search jobs	Must	Filters available for date range, operative, location and status; results update interactively.
FR-QS-04	Export cost reports	Should	Export filtered list as CSV or PDF preserving headings, totals and metadata.
FR-QS-05	Approve or flag jobs for review	Should	QS can set status {Approved, Flagged, Pending} with optional comment; status visible to operative.
FR-QS-06	View summary analytics	Could	Dashboard tiles: total cost (period), average cost per job, top cost drivers, week-on-week trend.

2.3 Cost Calculation Engine

Costs are computed server-side from a versioned rate card so that the same inputs always yield the same outputs, and historical jobs remain priced against the rates that were live when the work was done.

ID	Requirement	Priority	Acceptance Criteria
FR-CC-01	Compute labour cost from on-site duration	Must	labour = (departure – arrival in hours) × operative_hourly_rate; rounded to 2 dp.
FR-CC-02	Compute plant/tools cost	Must	plant = $\sum$ (rate_per_hour × duration_of_use_hours) for each selected item.
FR-CC-03	Compute materials cost	Must	materials = $\sum$ (unit_cost × quantity) across all material line items.
FR-CC-04	Compute vehicle cost	Must	vehicle = $\sum$ (daily_rate per vehicle) + optional mileage × per_mile_rate.
FR-CC-05	Compute grand total	Must	total = labour + plant + materials + vehicle, presented to QS and stored with the record.

ID	Requirement	Priority	Acceptance Criteria
FR-CC-06	Rate cards are versioned and date-bound	Should	Each cost calculation references the rate-card version active at job date, not the latest.

3. Non-Functional Requirements

Non-functional requirements describe how the system must behave: the qualities, constraints and standards that make Backpack fit for the working conditions of a regional electrical contractor and the regulatory environment of public-sector framework contracts.

ID	Requirement	Priority	Acceptance Criteria
NFR-01	Mobile-first interface designed for on-site use	Must	Primary controls reachable with one thumb; tap targets $\geq 48\text{dp}$; high-contrast palette readable in sunlight.
NFR-02	Responsive web dashboard for QS users	Must	Layout adapts cleanly between 1280px desktop and 1024px tablet; no horizontal scroll at supported widths.
NFR-03	Secure authentication with role-based access control	Must	Credentials hashed; sessions expire; routes guarded by role; failed sign-in attempts rate-limited.
NFR-04	Data encrypted in transit and at rest	Must	TLS 1.2+ for all traffic; database and file storage encryption enabled at provider level.
NFR-05	Offline-first architecture with reliable sync	Must	All capture flows work offline; queued records sync exactly once; user notified of unresolved conflicts.
NFR-06	Accessibility — WCAG 2.1 AA compliance	Should	All interactive elements have labels; colour contrast $\geq 4.5:1$; keyboard navigation supported on web.
NFR-07	Performance — perceived load under two seconds	Should	Mobile screen transitions $< 300\text{ms}$; dashboard list view $< 2\text{s}$ on broadband at 1,000 records.
NFR-08	Scalability across J McCann's asset base	Should	System designed to handle 62,000+ assets and a projected 250+ concurrent operative sessions.
NFR-09	Auditability of every cost record	Should	Each record stores who, when, rate-card version and any subsequent edits, immutably.
NFR-10	Maintainability and clean architecture	Should	Code passes flutter analyze with zero errors; $\geq 60\%$ unit-test coverage on cost engine.

4. Constraints & Assumptions

4.1 Constraints

- Delivery window: prototype completed and submitted by 17:00 on 1 June 2026.
- Technology: mobile and web client built in Flutter; backend hosted on Firebase to fit the timescale.
- Cost rates used in the prototype are illustrative mock values, not J McCann commercial data.
- Single-language interface (English) within the prototype scope.

4.2 Assumptions

- Operatives have company-issued Android or iOS devices with cameras.
- QS users access the dashboard via a modern desktop browser (Chrome, Edge, Safari, Firefox).
- Rate cards will ultimately be maintained by office staff in a separate admin interface, out of scope here.
- Site connectivity is intermittent but periodic sync windows are available daily.

5. Traceability Summary

A snapshot of requirement coverage across the prototype. Each identifier above can be mapped forward to the UI sketch, the coded screen and the cost-calculation tests delivered alongside this document.

Category	Count	ID Range
Operative — Mobile App	12	FR-OP-01 to FR-OP-12
QS — Web Dashboard	6	FR-QS-01 to FR-QS-06
Cost Calculation Engine	6	FR-CC-01 to FR-CC-06
Non-Functional Requirements	10	NFR-01 to NFR-10
Total	34	—

End of document.